Author Declarations and Transparency Statements

The Epistemic Fluency Trap in AI-Augmented Security Operations: Generative Rationale Fluency, Contested Evidence Cues, and Analyst Verification

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Data and Code Availability Statement

All source code, the `SOC-ITPI-24` benchmark, multi-corpus evaluation scripts, empirical log-likelihood lookup tables, and trace-driven decision-theoretic sensitivity pipelines supporting the findings of this study are openly available in an anonymous repository for double-blind peer review at https://anonymous.4open.science/r/epistemic-fluency-trap/ (camera-ready archival repository: https://github.com/tranducle/epistemic-fluency-trap). The underlying telemetry records are derived from four public cybersecurity repositories (`OTRF Security-Datasets`, `SALAD-SOC`, `LMD-2023`, and `CISA KEV`), with all prompt-injection strings sanitized using non-weaponizable administrative text.

Ethics Statement

This study did not involve any human participants, human-subject experiments, surveys, interviews, or animal subjects. All empirical evaluations and trace-driven decision-theoretic analyses were conducted exclusively on publicly available, de-identified cybersecurity telemetry datasets (`OTRF Security-Datasets`, `SALAD-SOC`, `LMD-2023`, and `CISA KEV`). Accordingly, institutional review board (IRB) human-subjects approval and informed consent were not applicable.

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Conflict of Interest Disclosure

The author declares no conflicts of interest.

CRediT Author Contributions and Acknowledgments

Tran Duc Le: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project Administration.

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Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the author used generative AI and AI-assisted tools strictly for grammar checking, academic tone polishing, LaTeX formatting, drawing LaTeX/TikZ figures, and code review. After using these tools, the author reviewed, verified, and edited all content and code as needed and takes full responsibility for the integrity, accuracy, and scientific claims of the publication.